เลขที่นั่งสอบ

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Midterm Examination of Semester 1 Year: 2017

Subject: 341151 Electric Circuits I Section: 05-06

Date: 5 October 2017 Time: 15.00-17.00

Name:____________________________ID:___________Field of Study:_________________

Instructions:

1. The examination has 11 pages (including this page), 3 sections (49 questions) and a total score of 65 points.
2. Write all your solutions and answers on this examination sheet.
3. This is a closed book examination.
4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
6. You are not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculators is allowed in the examination.
9. Electronic communication devices are NOT allowed in the examination room.
10. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

Part 1 (15 points) True/False

Instruction: Mark the correct answer for following questions in the provided answer sheet.

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1. 1,000,000 ohms is 10×10^{6}.
2. The prefix that corresponds to a power of ten of 9 is Giga.
3. Every source of voltage is created by simply making a separation of positive and negative charges.
4. A potential difference of 1 volt (V) exists between two points if 1 joule (J) of energy is exchanged in moving 1 coulomb (C) of charge between the two points.
5. Generally, Notations for sources is E (volts).
6. Electrical currents of 50 mA through the human body can cause severe shock.
7. Batteries consists of two or more similar cells. Primary type is not rechargeable.
8. From Coulomb's law $F = k\frac{Q_1 Q_2}{r^2}$, where k denotes the speed of light (3×10^{8}).
9. Kirchhoff's current law requires that total current input to a system must equal the total current output from the system.
10. The efficiency of a system is the ratio of the power input divided by the power output.
11. Following the Ohm's law, the more the resistance, the steeper the slope of the curve, obtained the relation between voltage and current.
12. The kilowatthour meter is used to measure current.
13. The ground fault circuit interrupt (GFCI) is designed to trip slower than the standard circuit breaker.
14. If I have a 6 V batter, a 12 V battery will work twice as well.
15. Most digital meters (DMM) have a fixed internal resistance of the voltmeter, which is in mega ohm range.

Part 2 (30 points) Multiple choices

Instruction: Mark the correct answer for following questions in the provided answer sheet.

1. Potential is defined as the _________ at a point with respect to another point in the electrical system.

   (a) charge (b) voltage

   (c) ampere (d) current

2. The voltage divider rule states that for two series ______________ of equal value, the voltage will divide equally.

   (a) elements (b) resistors

   (c) circuits (d) branches

3. If a short circuit carries a current of a level determined by the external circuit, the potential difference across its terminals _____________________.

   (a) cannot be measured (b) is always zero volts

   (c) may be zero volts (d) is always bigger than zero volts

4. Evaluate the following expression: $(15 \times 10^{-6})/(5 \times 10^{-9}) =$

   (a) 3×10^{-3} (b) 3×10^{3}

   (c) 0.03 (d) 300

5. In double-subscript notation, the voltage V_{ab} is the same as

   (a) $V_a - V_b$ (b) $V_a + V_b$

   (c) $V_a \times V_b$ (d) V_a

6. Four 12-volt batteries connected together by a conductor, positive terminal to negative terminal will result in which of the following voltage?

   (a) 3 volts (b) 6 volts

   (c) 12 volts (d) 48 volts

7. See Figure 1. The total resistance in this circuit is

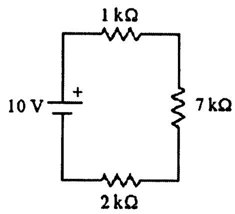


Figure 1.

(a) 1 kΩ (b) 2 kΩ

(c) 7 kΩ (d) 10 kΩ

8. See Figure 1. The total current flowing from the battery is

(a) 1 mA (b) 10 mA

(c) 1.43 mA (d) 5 mA

9. See Figure 1. The total power dissipated by the 7 kΩ resistor is

(a) 1 mW (b) 2 mW

(c) 7 mW (d) 10 mW

10. The sum of the series voltage drops around a closed loop must_________.

(a) equal infinity (b) equal the applied voltage

(c) equal 120 volts (d) be the same

11.See Figure 2. Which one of the following KVL equations describes this circuit?

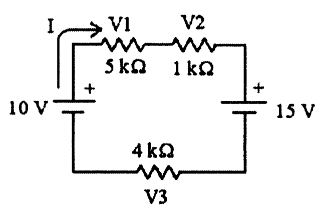


Figure 2.

(a) +10 V + V1 + V2 - 15 V + V3 =0

(b) +10 V - V1 - V2 - 15 V - V3 = 0

(c) -10 V - V1 - V2 + 15 V - V3 =0

(d) +10 V - V1 - V2 + 15 V - V3 = 0

12.See Figure 2. The total current I is

(a) +500 μA (b) -500 μA

(c) +2.5 mA (d) -2.5 mA

13.A circuit has a source voltage of 15 volts and two resistors with a total resistance of 4000 ohms. If R1 has a potential of 9.375 volts, what is the value of R2?

(a) 500 ohms (b) 1000 ohms

(c) 1500 ohms (d) 2000 ohms

14.See Figure 3. What is the current I_2?

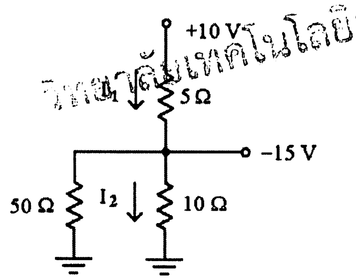


Figure 3.

(a) -1.5 A (b) +1.5 A

(c) -0.67 A (d) +0.67 A

15.See Figure 3. What is the current I_1?

(a) +0.75 A (b) +2.0 A

(c) -5.0 A (d) +5.0 A

16.The attraction force between two charges which can be determined by__________.

(a) Coulomb's law (b) Newton's law (c) Kirchhoff's current law

(d) Ohm's Law

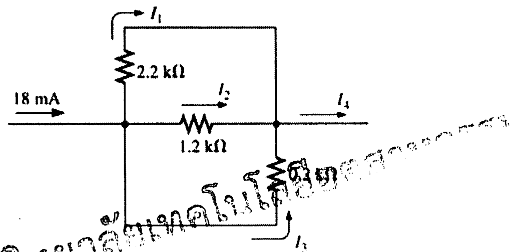
17.How much charge passes through a battery of 4.5 V if the energy expended is 60 J?

(a) 13.5 C (b) 13.3 C (c) 10.6 C (d) 11.4 C

18.What is the total cost of using an 1,000 W air conditioner for 12 h at a rate of 10 Baht per kilowatthour?

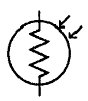
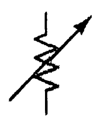
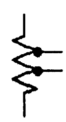
(a) 160 Baht (b) 120 Baht (c) 12.0 Baht (d) 240Baht

19.Following the circuit below, find the current I_4.

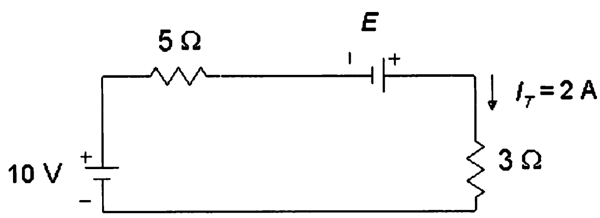


(a) 18.3 mA (b) 1.8 mA (c) 18 mA (d) 18.5 mA

20.What is the symbol of Photoresistor?

(a) 
(b) 
(c) 
(d) 

21.Find the value for the unknown voltage source.



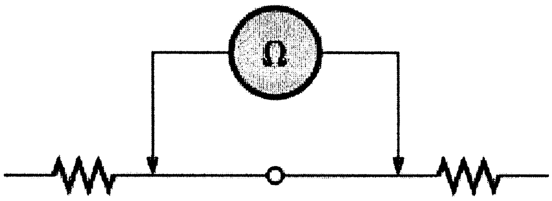
(a) 8 V (b) 6 V (c) 4 V (d) 2 V

22.Find the value of R in picture below.

A resistor colored Green-Red-Gold-Silver

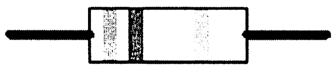
(a) 9 Ω (b) 5.2 Ω (c) 1.8 Ω (d) 52 Ω

23.Find the value of resistance in picture below.



(a) 0 Ω (b) 0.1 Ω (c) 0.001 Ω (d) 10 Ω

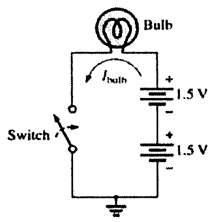
24.Find the dissipated power of R in picture below, where the current is 1A.



A resistor colored Green-Red-Gold-Silver

(a) 9 w (b) 5.2 w (c) 1.8 w (d) 52 w

25.Determine the voltage across the bulb when the switch is closed.



(a) 3 V (b) 3.5 V (c) 0.7 V (d) 1.5 V

26.What is the meaning of 5W resistor.

5W

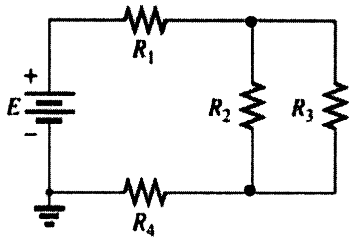
(a) 5W of dissipated power.

(b) 5W of maximum power handling.

(c) 5W power is converted to the light.

(d) 5W of minimum power handling.

27.Find the value of R_1. Given R_2=2Ω, R_3=2Ω, R_4=5Ω and R_T=10Ω.

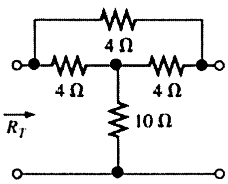


(a) 9 Ω (b) 4 Ω (c) 8 Ω (d) 20 Ω

28.How large is dissipated power (W) of a 5 Ω resistor connected across a 10 V battery?

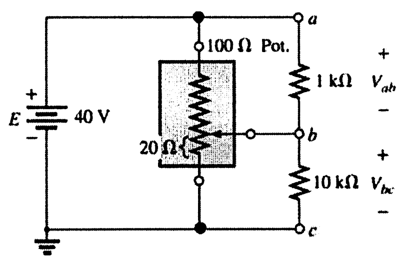
(a) 20W (b) 2W (c) 10W (d) 1W

29.Calculate R_T.



(a) 14 Ω (b) 12.6 Ω (c) 24 Ω (d) 18 Ω

30.What is the voltage V_{ac}?

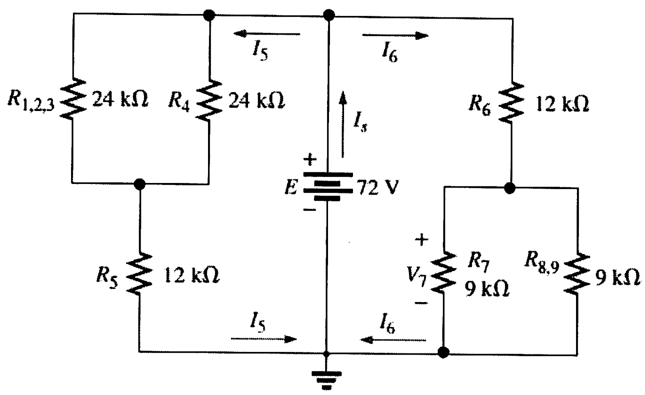


(a) 40 V (b) 3.63 V (c) 36.36V (d) 31.5 V

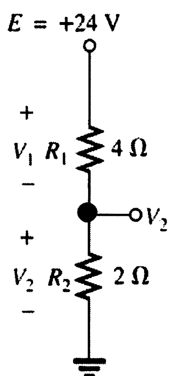
Part 3 (20 points) Calculation

Instruction: Show the mathematical expression and answers of following problems.

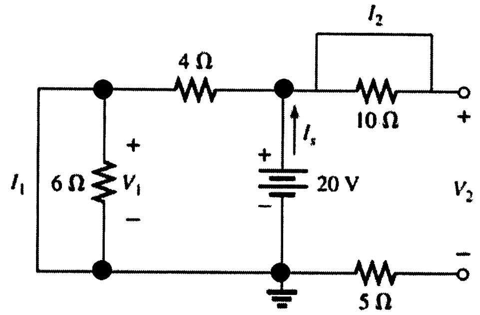
1. From Figure below, calculate R_T and I_S. (5 points)



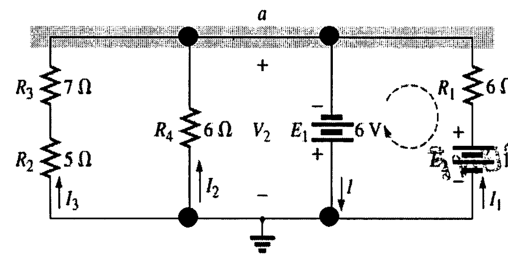
2. Use the voltage divider to find the value of V_2 . (5 points)



3. Determine I_s, I_1, I_2, V_1 and V_2 from the following circuit. (5 points)



4. Find current I, voltage V_1 and V_2. (5 points)



Answer Sheet

Name:______________________________ ID:______________

Subject: 341151 Electric Circuits I

Part 1

	True	False
1.		
2.		
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4.		
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Part 2

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